

Data warehouse: ERP and CRM to a star schema

A three-layer SQL Server warehouse that turns raw ERP and CRM exports into a documented star schema.

TYPE Personal project	TOOLS SQL Server, T-SQL, ETL	DATA 6 CSV files (CRM and ERP)
BUILT Feb 2025		

01

The question

Two source systems, a CRM and an ERP, export raw files in different shapes. How do you combine them into one clean model that analysts can query?

02

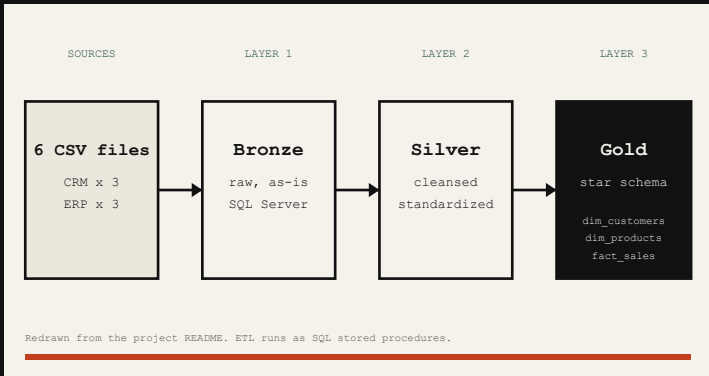
The data

Six CSV files: three from the CRM (customer info, product info, sales details) and three from the ERP (customer, location and product-category tables). The scope is the latest data only; history is not tracked.

03

The method

A medallion architecture in SQL Server, built in three layers.



The three layers, redrawn from the project README.

- **Bronze:** raw files loaded as they arrive, by a stored procedure.
- **Silver:** cleansing and standardisation of both sources, also by stored procedure.
- **Gold:** business-ready views in a star schema: `dim_customers`, `dim_products` and `fact_sales`.

The repo also holds a column-level data catalog for the gold layer and a set of naming conventions.

04

What I found

This project is about building the model, not about a business finding. What it delivers is a single place where CRM and ERP data meet: one customer dimension, one product dimension and one sales fact table, documented well enough for someone else to query.

05

What I'd do next

- Add automated checks that row counts and totals reconcile between layers.
- Keep history, for example slowly changing customer attributes, rather than only the latest snapshot.
- Build a Power BI report on the gold layer for customer behaviour, product performance and sales trends.

06

Links

- [Scripts, data catalog and docs on GitHub](https://github.com/samriddhosaha/SQL-PortfolioProjects/tree/main/Data-Warehouse-Project) ↗ [\[https://github.com/samriddhosaha/SQL-PortfolioProjects/tree/main/Data-Warehouse-Project\]](https://github.com/samriddhosaha/SQL-PortfolioProjects/tree/main/Data-Warehouse-Project)

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